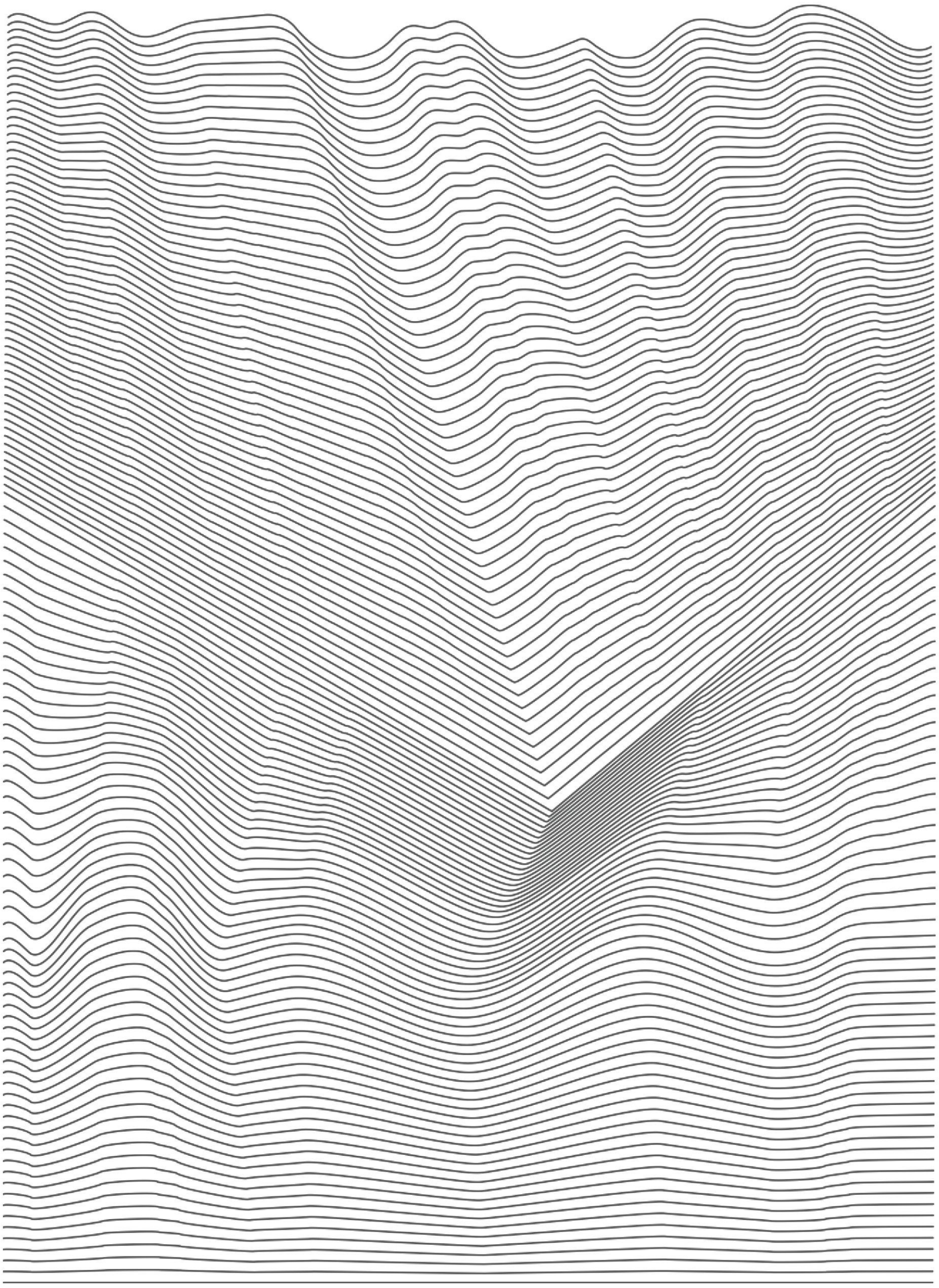
WORK SAMPLE



2020–2022

HUNTER KEENE

HUNTER KEENE

hunterjkeene@gmail.com

702-592-9301

hunterkeene.myportfolio.com

JOB EXPERIENCE

FREELANCE 2D/3D DRAFTSMAN *PRESENT*

PRIVATE CLIENTS IN NJ , PA, NY

WALKTHISHOUSE *2020 - PRESENT*

INTERNSHIP TO PART TIME POSITION

HOBOKEN NJ

NEW YORK CITY STAGEHAND *2018 - PRESENT*

INCLUDING THEATRICAL **RIGGING AND CARPENTRY**

LOCAL ONE IATSE AND UNDER PRIVATE CONTRACTING COMPANY

MARK GOULD ARCHITECTURE *2018*

INTERNSHIP

NEW YORK, NY

WORKED ON **SURVEYING AND DRAFTING**

EASTERN MILLWORKS, INC *2016*

INTERNSHIP

JERSEY CITY, NJ

ARCHITECTURAL WOODWORKING COMPANY

SCHOLARSHIPS AND AWARDS

DESIGN AND BUILD COMPETITION 2021

MUSEUM OF OUTDOOR ARTS- MONUMENT FOR MARS

HONORABLE MENTION

SCHOLASTIC ACHIEVEMENT AWARD 2021

AIA ARCHITECTS LEAGUE OF NORTHERN NEW JERSEY

SCHOLARSHIP AWARDED

EDUCATION

THE PENNSLYVANIA STATE UNIVERSITY

STATE COLLEGE, PA

PROGRAMS

AUTOCAD	●●●●
PHOTOSHOP	●●●●
ILLUSTRATOR	●●●●
RHINO	●●●○
SKETCHUP	●●●○
V RAY	●●●○
INVENTOR	●●○○
3DS MAX	●●○○
INDESIGN	●●○○
VECTORWORKS	●●○○
GRASSHOPPER	●○○○

SKILLS

LASER CUTTER	●●●●
3D PRINTING	●●●○
CARPENTRY	●●○○
CNC	●●○○
SEWING	●●○○
LOOM KNITTING	●●○○
SCREEN PRINTING	●●○○

2019 ALBERT O. HALSE AWARD

AIA ARCHITECTS LEAGUE OF NORTHERN NEW JERSEY

SCHOLARSHIP AWARDED

2018 PROVOST AWARD

SCHOLARSHIP AWARDED FROM **THE PENN STATE UNIVERSITY**

BACHELORS OF ARCHITECTURE

MINOR IN ARCHITECTURAL HISTORY

GRADUATED 2022

INDEX

PROJECTS

CONCRETE-BASED MEMBRANES 3

SMOCKED MATERIAL AS FORMWORK FOR PANEL SYSTEM

HIGH PERFORMANCE BUILDING 7

168,000 SQFT STEM HIGH SCHOOL WITH SUSTAINABLE GOALS

LIGHT WEIGHT TENSION STRUCTURE 9

FEEDBACK LOOPS BETWEEN KNITTED SURFACES AND SIMULATIONS

MEDIA CITY PERFORMANCE CENTER 11

TECHNICAL SYSTEM INTEGRATION BUILDING DESIGN

MONUMENT FOR MARS 13

2021 HONORABLE MENTION, MUSEUM OF OUTDOOR ART

INTERESTS

RESEARCH	THE EXPLORATION OF NEW TECHNIQUES AND METHODS
DESIGN-BUILD	CONCEPTUAL DESIGN TO BUILT CONDITIONS
DIGITAL-PHYSICAL FEEDBACK	INTERACTIONS BETWEEN DIGITAL SIMULATION AND PHYSICAL CONDITIONS
SUSTAINABILITY	PLACING EMPHASIS ON PASSIVE DESIGN STRATEGIES
MATERIAL	EXPLORING THE LIMITS OF MATERIALS AND THEIR EFFECT ON DESIGN CONDITIONS
FABRICATION	THE CRAFT AND EXECUTION OF CONSTRUCTION ON MANY SCALES

CONCRETE-BASED MEMBRANES

DIRECTED RESEARCH STUDIO FALL 2021 15 WEEKS

IN THIS STUDIO I RESEARCHED AND EXPLORED THE PROCESS OF NON RIGID FORMWORK FOR CREATING CONCRETE FORMS. I BEGAN BY EXPERIMENTING WITH A VARIETY OF MATERIALS AND MORTAR MIXTURES.

MATERIAL EXPLORATION

NON-RIGID FORMWORK EXPERIMENTS

#1



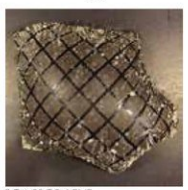
CANVAS; COTTON/ LINEN BLEND
6" x 6" x 2"
Δ DEPTH 0"
CONCRETE
UNTAILORED

#2



NYLON
6" x 6" x 3.25"
Δ DEPTH 1.5"
CONCRETE
PARACORD LATTICE RESTRAINT

#3



POLYPROPYLENE
6" x 7" x 2.75"
Δ DEPTH 0"
CONCRETE
UNTAILORED

#4



POLYPROPYLENE
6" x 7" x 2.75"
Δ DEPTH 0"
CONCRETE
UNTAILORED

#5



NYLON
6" x 6" x 2"
Δ DEPTH 2"
MORTAR
SEAM STITCHED INTO FABRIC

#6



CANVAS; COTTON/ LINEN BLEND
6" x 6" x 3.5"
Δ DEPTH 0"
MORTAR
SMOCKED

#7



WOVEN POLYPROPYLENE BAG
12" x 6" x 4"
Δ DEPTH 0"
CONCRETE
SMOCKED

#8



WOVEN POLYPROPYLENE BAG
12" x 6" x 4.5"
Δ DEPTH 0"
CONCRETE
SMOCKED

#9



1/16" LATEX RUBBER
6" x 6" x 2.75"
Δ DEPTH 1.5"
MORTAR
SMOCKED

#10



2 MIL LATEX RUBBER
6" x 6" x 4"
Δ DEPTH 4"
MORTAR
SMOCKED

RETENTION METHODS

POUR 12





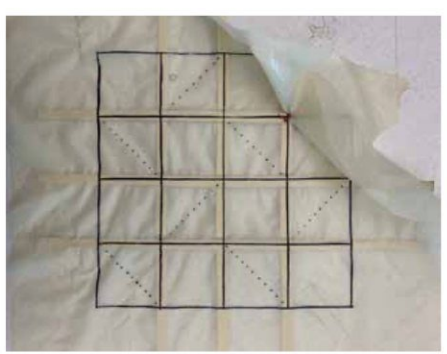


POUR 13





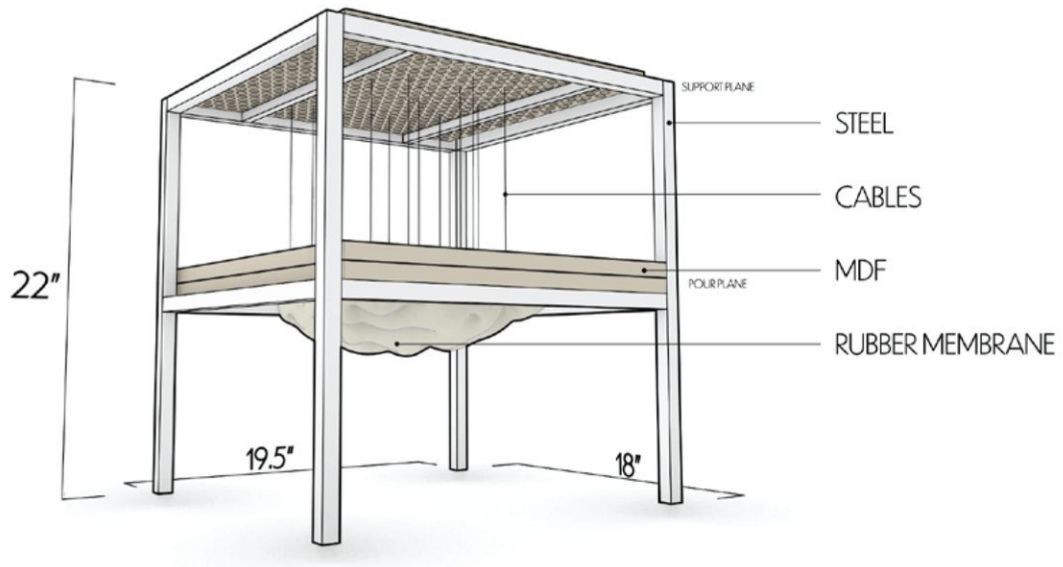




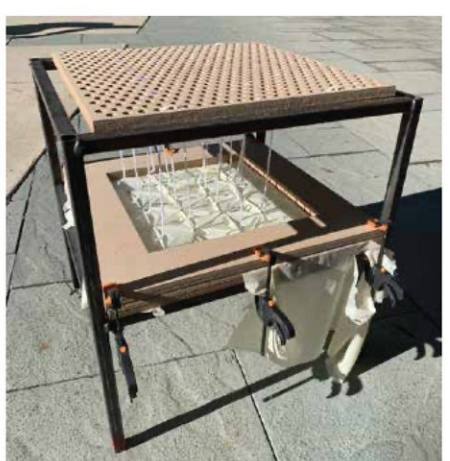
RESEARCH DESIGN-BUILD DIGITAL-PHYSICAL FEEDBACK SUSTAINABILITY MATERIAL FABRICATION

SYSTEM DEVELOPMENT

THE RIG Dual plane retention system for the production of folded facade panels

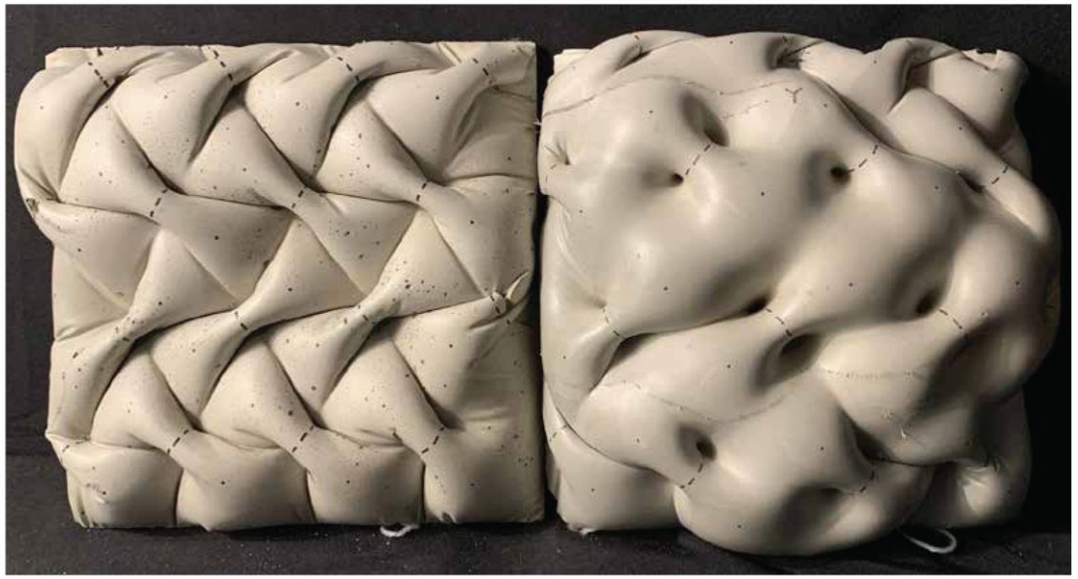


THE RIG WAS THE PRODUCT OF A NEED FOR CONTROL OVER THE EXPANSION OF THE RUBBER MEMBRANE UNDER THE WEIGHT OF THE MORTAR. IT ALLOWED FOR A HIGH LEVEL OF CONTROL. BY SUPPORTING EACH INDIVIDUAL STITCH THE OVERALL SHAPE OF THE FINAL MOLD COULD BE PRESCRIBED.



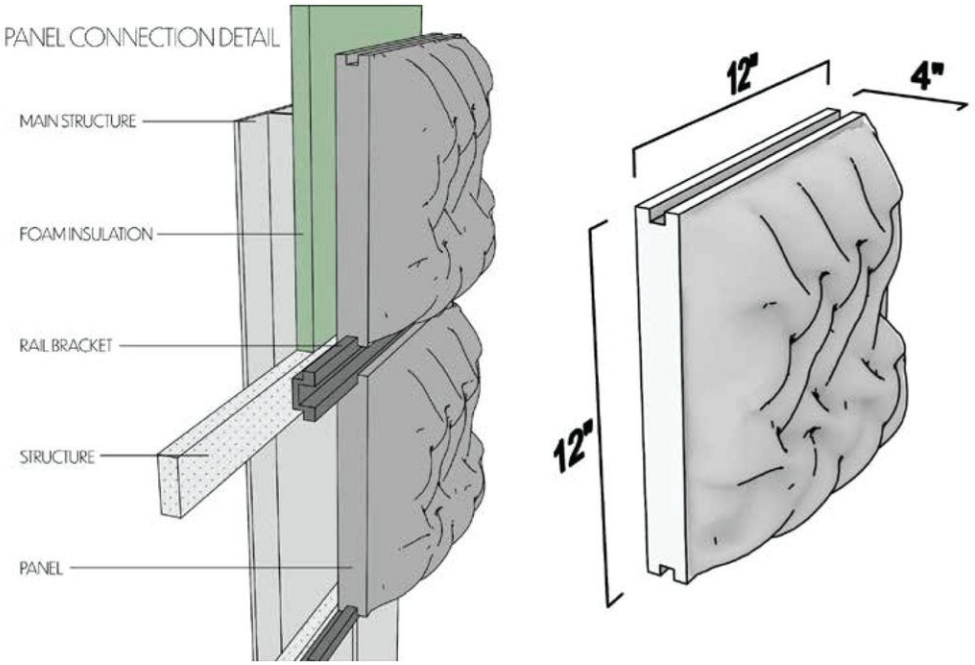


PANEL DEVELOPMENT

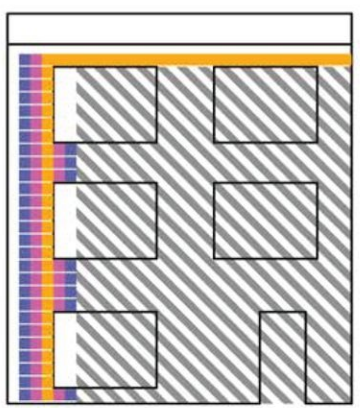


WITH A FOCUS ON DEVELOPING A VARIABLE FACADE SYSTEM THAT WOULD ALLOW FOR DESIGNED AREAS OF HIGH AND LOW THERMAL MASS AND INCREASED SHADING, THREE PANEL TYPES WERE CREATED.



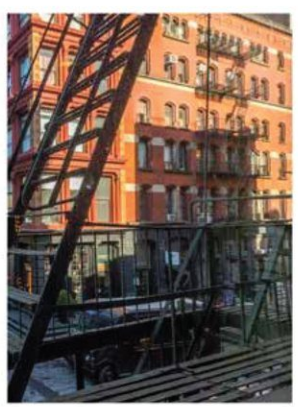


PANELS AS AN INTERVENTION TO AN EXISTING FACADE



A dense urban setting can create inconsistent sun conditions

Panels can be arranged on facade to utilize this solar energy and to regulate the heating/cooling load

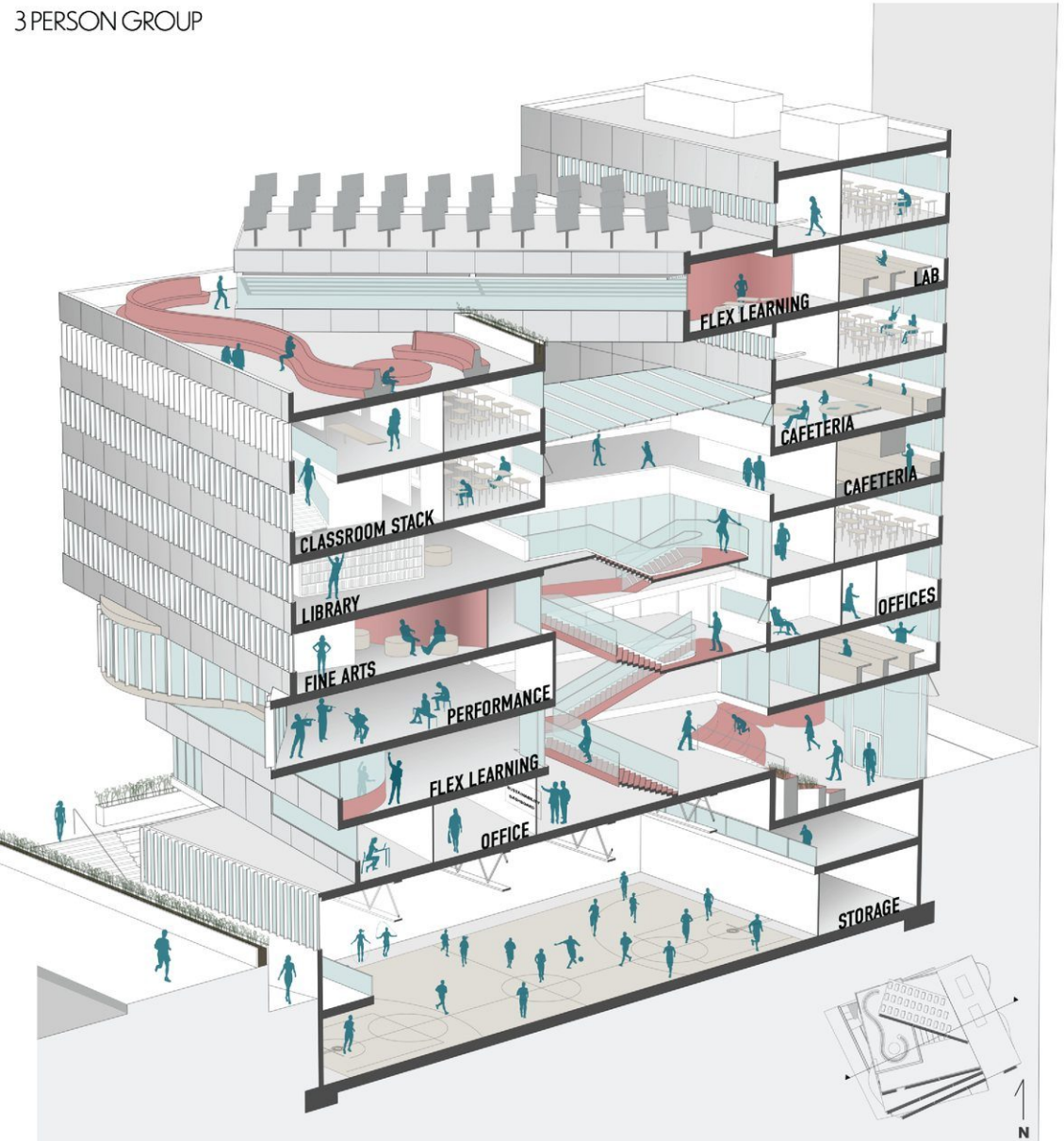




HIGH PERFORMANCE BUILDING

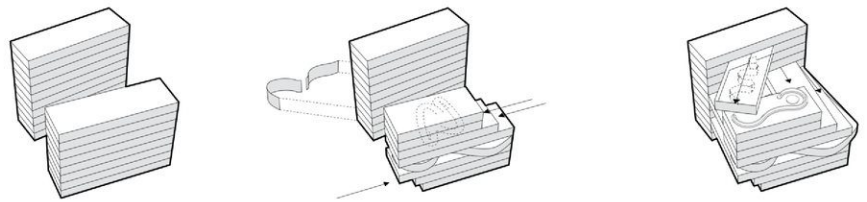
DIRECTED RESEARCH STUDIO | FALL 2020 | 15 WEEKS

3 PERSON GROUP



THIS STUDIO FOCUSED ON SUSTAINABILITY. WE UTILIZED ANALYTICAL PROGRAMS SUCH AS COVE TOOL TO IMPROVE THE PERFORMANCE OF THE BUILDING. THE PROGRAM CONSISTED OF A 168,000 SQF STEM HIGH SCHOOL LOCATED IN PITTSBURGH, PA. THIS PROJECT SERVED AS OUR ENTRY TO THE COTE TOP 10 COMPETITION WHICH WAS IMPLEMENTED INTO OUR CURRICULUM.

FORM GENERATION AND CONCEPT



THIS STRATEGY INTENDS TO FOSTER A CLEAR INTERDEPENDENCE BETWEEN STEM EDUCATION AND CREATIVITY WITH THE IDEA THAT ONE CANNOT PROSPER AND GROW WITHOUT THE OTHER.

PERFORMANCE FEATURES

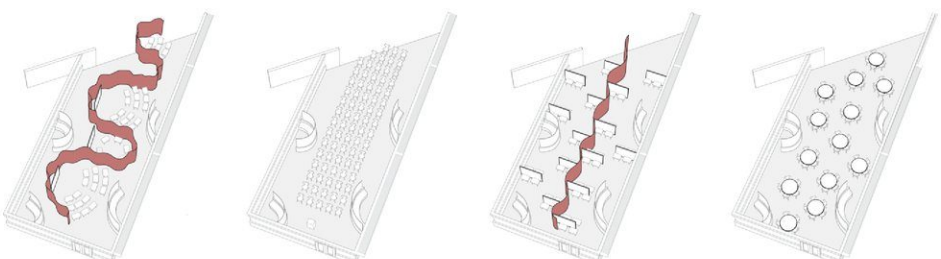
Adaptable Design


VARIETY OF SPACIAL STRUCTURES


MOVABLE PARTITION WALLS


FLEXIBLE LARGE SPAN

Flexible Multi-Use Spaces



LEARNING	LECTURING	TESTING	STUDYING
Curtain track #1 divides spaces and allows for a variety of class sizes	Curtain retracted creates a large meeting space	Curtain track #2 creates moderate divisions of space and rolling partitions provide privacy	Curtain retracted with no division of spaces allows for tables which provide communal study areas

OUTDOOR SPACES

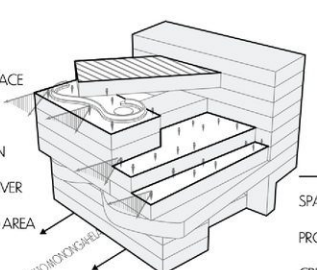
OFFSET MASSES CREATE TERRACED OUTDOOR SPACE

MAIN AREAS OF BUILDING HAVE DIRECT ACCESS

PROMOTES INTERDISCIPLINARY COMMUNICATION

FRAMES VIEW OF ADJACENT MONONGAHELA RIVER

STRENGTHENS CONNECTION TO SURROUNDING AREA



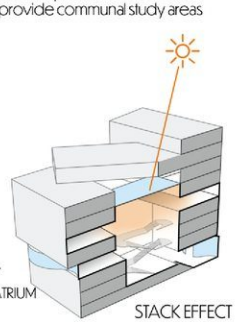
INTEGRATED ATRIUM

SPACES ORGANIZED AROUND THE CENTRAL ATRIUM

PROVIDES PASSIVE HEATING DURING WINTER

CREATES PASSIVE VENTILATION THROUGH ATRIUM

SHARED CENTRAL CIRCULATION PROMOTES COMMUNICATION

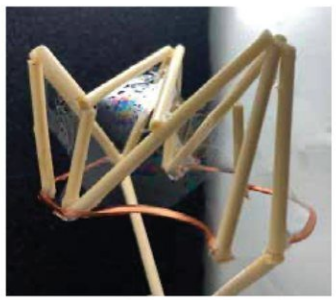
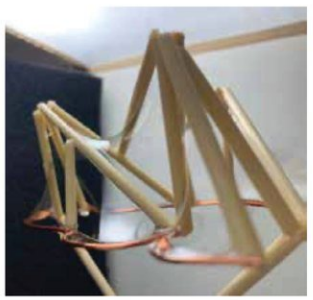
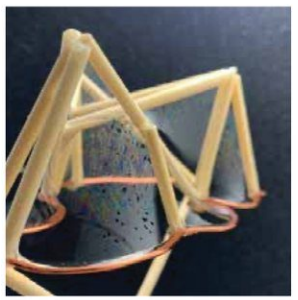

STACK EFFECT

LIGHT WEIGHT TENSION STRUCTURE

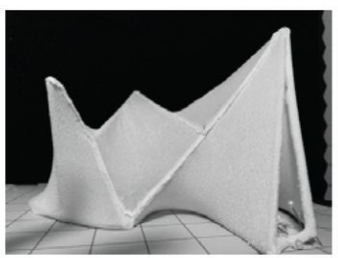
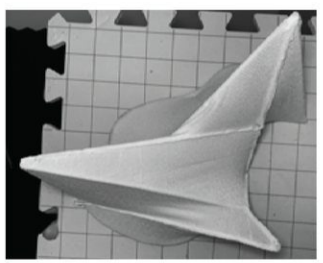
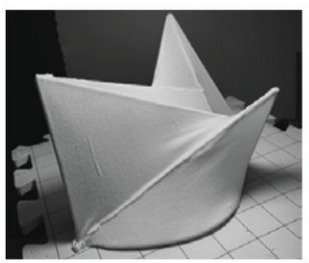
DIRECTED RESEARCH STUDIO SPRING 2021 15 WEEKS
3 PERSON GROUP

TENSION SURFACE EXPLORATIONS - HYPERBOLIC PARABOLOIDS

PHILLIPS PAVILION

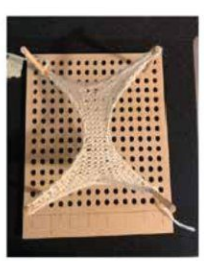
WIRE BAMBOO SKEWERS 1: 2 PUSTEFIX TO WATER

WIRE BAMBOO SKEWERS NYLON STOCKINGS

KNITTED HY-PAR SERIES WITH DIGITAL SIMULATION

THIS STUDIO EXPLORED THE RELATIONSHIP BETWEEN PREDICTIVE GRASSHOPPER ALGORITHMS AND PHYSICALLY KNITTED TEXTILES. STARTING WITH PRECEDENT ANALYSIS, I FOCUSED ON THE HYPERBOLIC PARABOLOID AND ITS MANIFESTATION IN A VARIETY OF MEDIUMS.







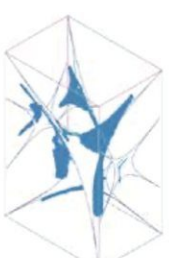
RESEARCH DESIGN-BUILD DIGITAL-PHYSICAL FEEDBACK MATERIAL FABRICATION



6' x 4' x 2.5'

USING GRASSHOPPER PLUGINS LIKE KANGAROO AND KIWI 3D I WAS ABLE TO TUNE MY KNITTING AND TECHNIQUES TO CREATE HOMOGENEOUSLY TENSIONED TEXTILES. THIS PROCESS WAS THEN SCALED UP FOR THE FINAL PROJECT.

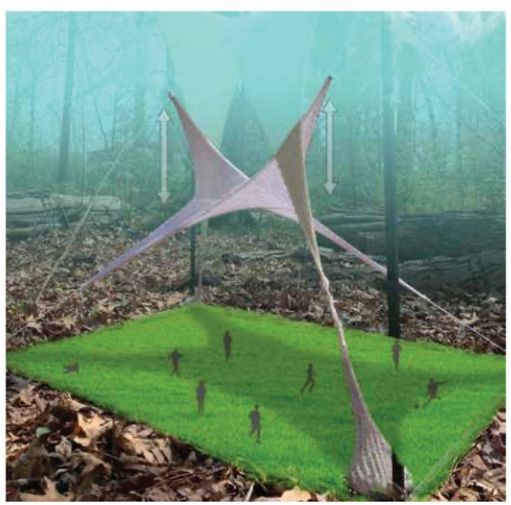




DIGITAL SIMULATION MERGED WITH 3D SCANS OF KNITS

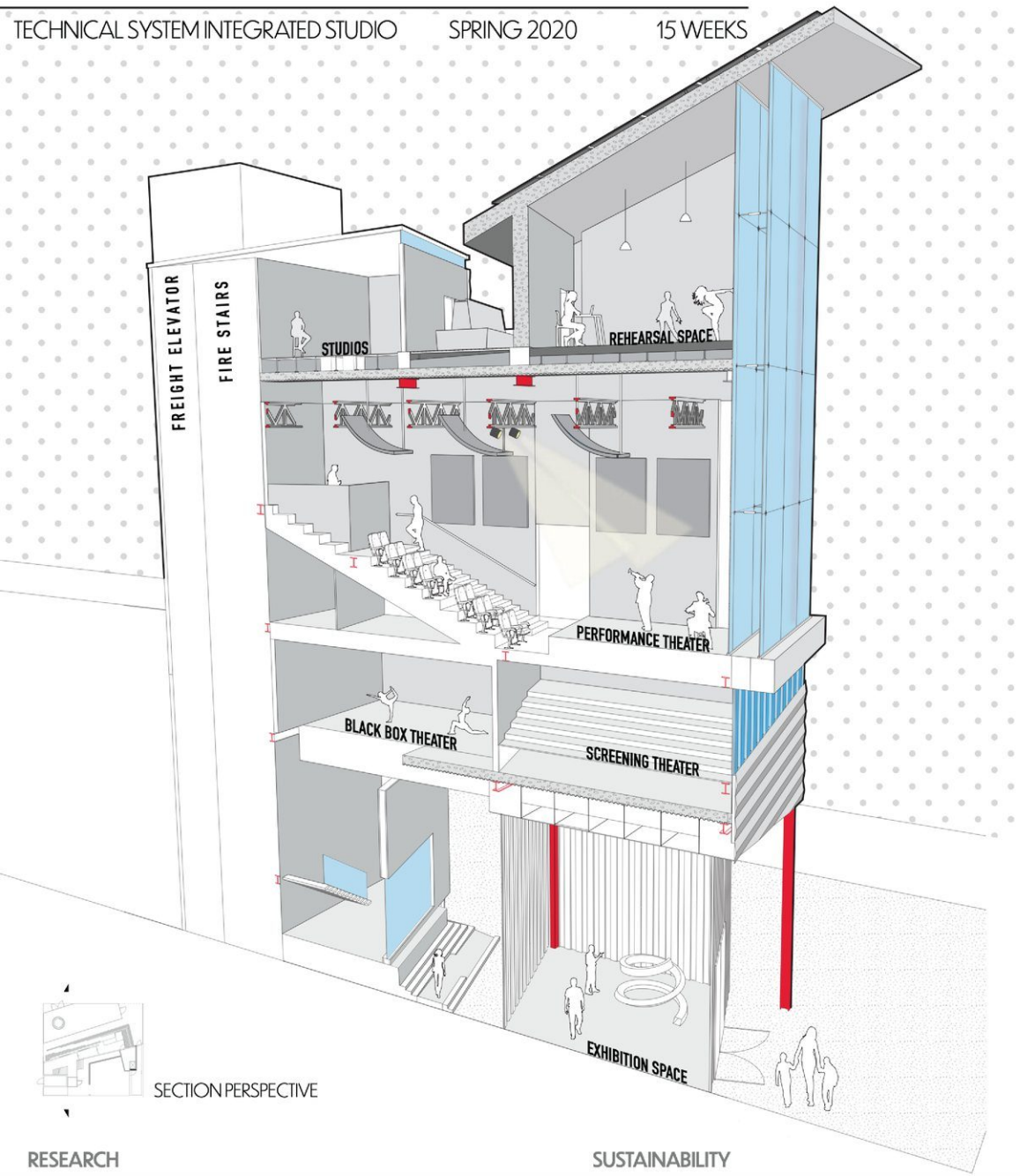


THE FINAL PRODUCT CONSISTED OF A VARIETY OF HAND KNITTED SHAPES. I WAS SELECTED TO KNIT THE LARGE HY-PAR AND TWO OF THE ACCOMPANYING SMALLER SHAPES. EACH SHAPE IS RELIANT ON THE TENSION OF THE OTHER CREATING A BALANCED TENSION CONDITION.



MEDIA CITY NOMINATED FOR DESIGN EXCELLENCE AWARD

TECHNICAL SYSTEM INTEGRATED STUDIO SPRING 2020 15 WEEKS



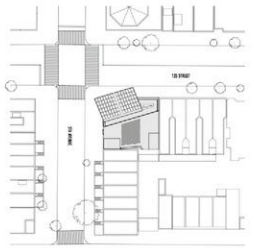
SECTION PERSPECTIVE

RESEARCH

SUSTAINABILITY

COMMUNICATION OF FUNCTION WITH COMMUNITY

THE OPPORTUNITY



125 AND 5 AVE, HARLEM NYC

A CORNER SITE CONDITION

HIGHLY EXPOSED URBAN CONDITION

REACTION TO ISSUE



SPACE IS GIVEN BACK TO PUBLIC

A "WALK-THROUGH" ALLOWS FOR PEOPLE TO EXPERIENCE BUILDING WITHOUT ENTERING

CORNER AS MEDIUM

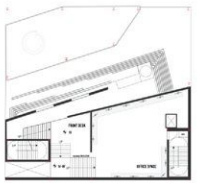


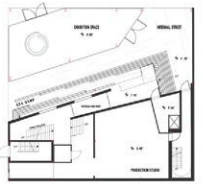
CORNER USED AS EXHIBITION SPACE

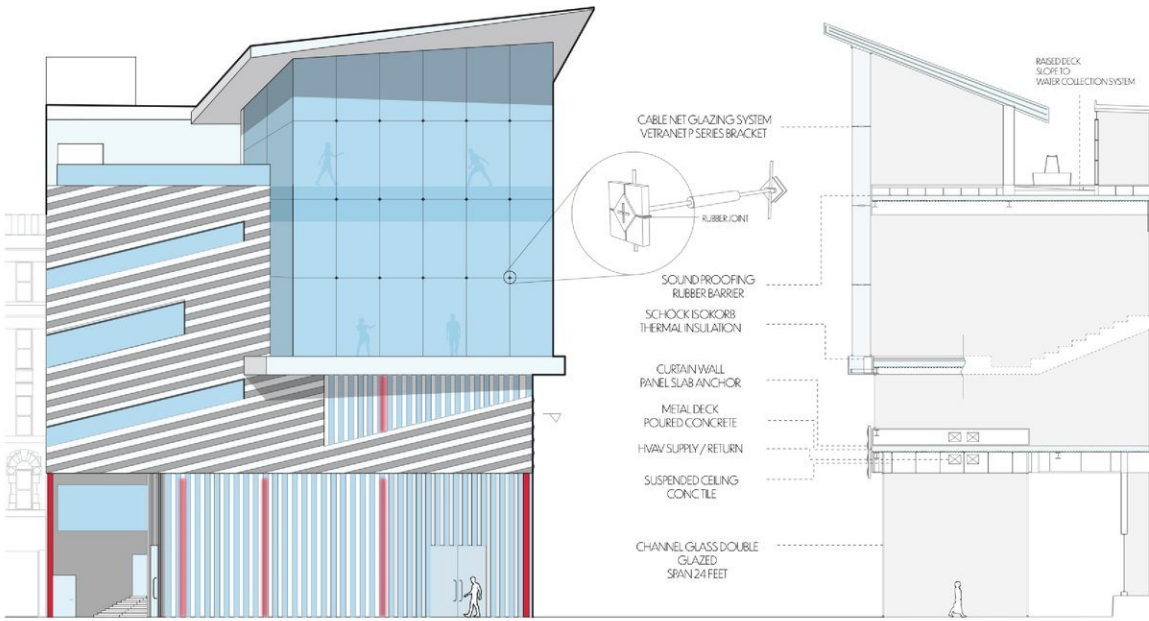
CORNER USED TO EXPOSE PROGRAM









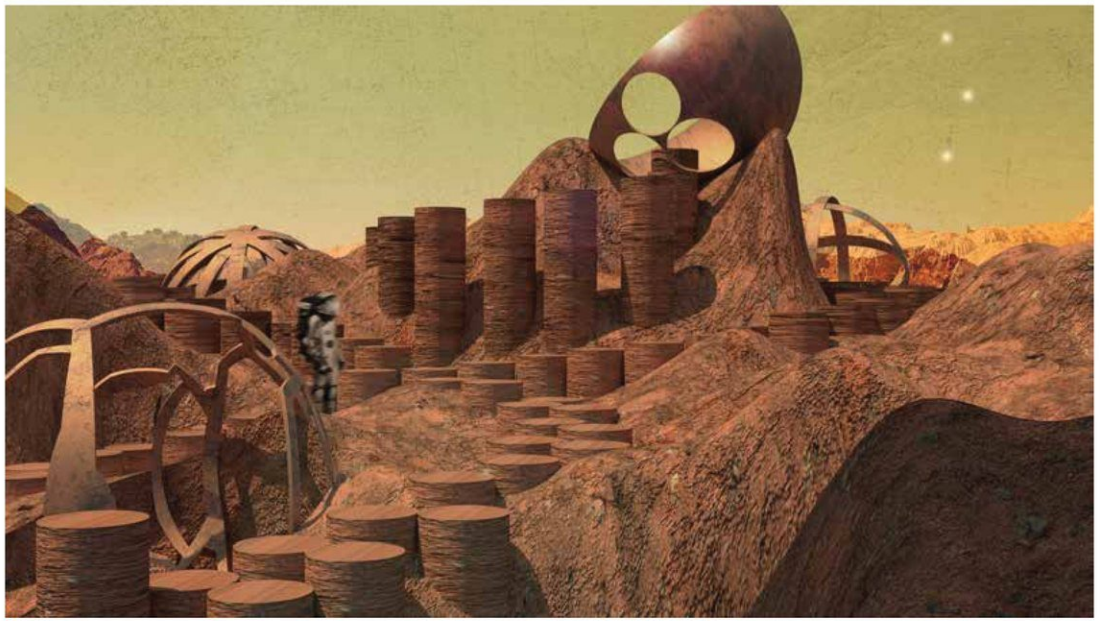


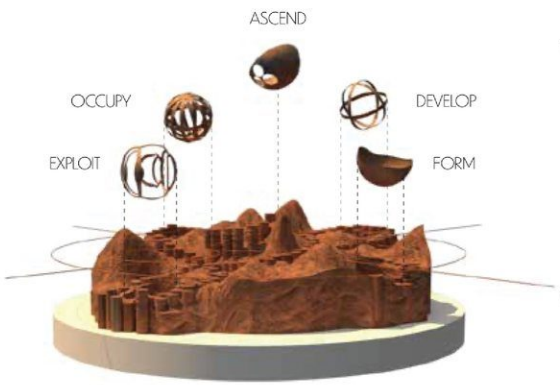
MONUMENT FOR MARS HONORABLE MENTION

INDEPENDENT COMPETITION ENTRY FALL 2022 6 WEEKS
2 PERSON GROUP

ASCEND - A CRITICAL MONUMENT

PROMPTED WITH DESIGNING A MONUMENT TO HUMANITIES TRANSITION TO LIVING ON MARS, ASCEND POINTS A CRITICAL LENS ON OUR INTERACTION WITH THE ENVIRONMENT. COMPOSED OF A SERIES OF SCULPTURAL MOMENTS FRAMED BY A WINDING PATH ; THE INDIVIDUAL STOOLS ARE FORMED OUT OF 3D PRINTED MARTIAN SOIL.

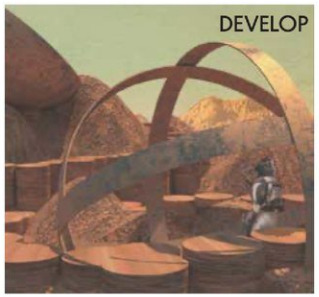




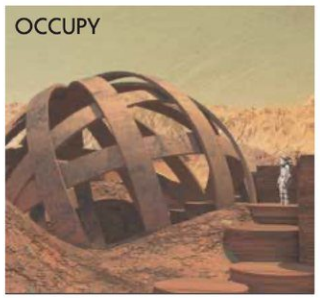
THIS MONUMENT AIMS TO QUESTION THE NEED FOR A TRANSITION TO MARS. WHILE IT'S AN EVENT TO BE CELEBRATED, THE CONDITION OUR EARTH IS CURRENTLY IN IS NOTHING TO FORGET ABOUT. THIS JOURNEY DOCUMENTS THE PROCESS IN WHICH OUR CULTURE **FORMED**, **DEVELOPED**, **OCCUPIED**, AND **EXPLOITED** OUR RESOURCES. THIS FRAMES THE **ASCENSION** TO A NEW PLANET AS A REGRETTABLE BUT NECESSARY VENTURE.

RESEARCH SUSTAINABILITY MATERIAL

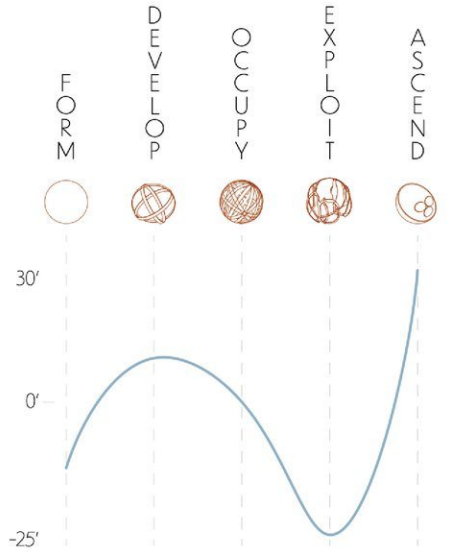




STEPS OF THE JOURNEY



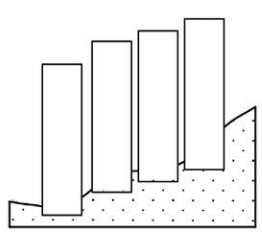




THE HEIGHT OF THE PATH IS BASED ON THE FLUCTUATION OF HUMANITY'S INTERACTION WITH OUR PLANET.
THE HIGHS AND LOWS OF OUR CULTURAL EVOLUTION ARE MIRRORED BY THE PEAKS AND VALLEYS OF THE PATH.

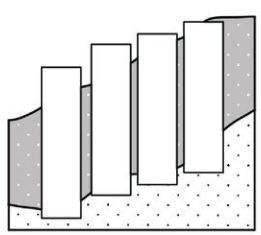
5 DISTINCT MOMENTS OUTLINE HUMANITIES INTERACTION WITH THE ENVIRONMENT. THE USER NAVIGATES THE HIGHS AND LOWS CREATED BY THE UNDULATING STOOLS. AS OPPOSED TO DESIGNING SOMETHING INTENDED TO PERMANENTLY MARK THE LANDSCAPE, ASCEND IS DESIGNED TO RETURN BACK TO THE EARTH FROM WHICH IT CAME.

MONUMENT LIFE CYCLE



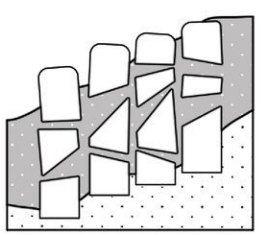
PHASE 1

THE INITIAL STATE OF THE MONUMENT.
MAJORITY EXPOSED AND IN NEW CONDITION.



PHASE 2

AS THE SANDS AND WIND SWEEP ACROSS THE MONUMENT THE GAPS IN BETWEEN THE STOOLS IS FILLED



PHASE 3

THE FINAL STAGE OF THE MONUMENT LEAVES MOST OF THE STOOLS COVERED IN SAND.
THE OBJECTS DEGRADE OVERTIME AND RETURN BACK TO THE GROUND FROM WHICH THEY CAME

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